

## Code-Layout, Readability and Reusability

|                      |                                        |
|----------------------|----------------------------------------|
| <b>Date</b>          | 10 July 2026                           |
| <b>Team ID</b>       | XXXXXX                                 |
| <b>Project Name</b>  | Rising waters- Flood prediction System |
| <b>Maximum Marks</b> | 5 Marks                                |

### Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

### Code Layout Checklist:

| S.No | Code Quality Parameter    | Description                                                           | Followed (Yes / No / Partial) | Remarks                                                |
|------|---------------------------|-----------------------------------------------------------------------|-------------------------------|--------------------------------------------------------|
| 1    | Consistent Indentation    | Uniform spacing/tabs used throughout the code                         | yes                           | Uniform indentation maintained throughout the project. |
| 2    | Proper File Structure     | Files and folders are logically organized logically for easy access   | Yes                           | Project files are well organized.                      |
| 3    | Meaningful Variable Names | Variables reflect their purpose clearly<br>Describe heir purpose      | Yes                           | Descriptive variable names are used.                   |
| 4    | Function / Method Names   | Functions are descriptively named<br>According to their functionality | Yes                           | Function names are meaningful and easy to understand.  |
| 5    | Code Comments             | Important code sections include functions and modules                 | Yes                           | Comments are provided where necessary.                 |
| 6    | Modular Design            | Code is divided into reusable functions and modules                   | Yes                           | Modular design improves maintainability.               |
| 7    | No Redundant Code         | Duplicate or unnecessary code has been removed                        | Yes                           | No redundant code is present.                          |
| 8    | Error Handling            | Exceptions and errors are handled properly                            | Yes                           | Basic exception handling is implemented.               |

### Reusable Components / Modules:

| S.No | Component / Module Name | Language / Technology | Where Reused                       | Reusability Level (High / Medium / Low) |
|------|-------------------------|-----------------------|------------------------------------|-----------------------------------------|
| 1    | Data preprocessing      | Python, Pandas        | Data Cleaning & data Preprocessing | High                                    |

|   |                        |                     |                            |        |
|---|------------------------|---------------------|----------------------------|--------|
| 2 | Feature Engineering    | Python, Numpy       | Model Training             | High   |
| 3 | Machine Learning model | Scikit-learn        | Flood Prediction           | High   |
| 4 | Data Visualization     | Matplotlib, seaborn | Data Analysis & Reports    | Medium |
| 5 | Prediction Module      | Flask, Python       | Web Application Prediction | High   |

**Overall Code Quality Assessment:**

| Aspect                   | Rating (1-5) | Comments                                               |
|--------------------------|--------------|--------------------------------------------------------|
| Code Layout & Structure  | 5            | Well-organized and modular code structure.             |
| Readability              | 5            | Clear variable names and easy-to-understand code.      |
| Reusability              | 5            | Components can be reused in future projects.           |
| Documentation / Comments | 5            | Sufficient comments and documentation provided.        |
| Overall Score            | 5            | High-quality, maintainable, and reusable project code. |